

Name: _____

Class: _____

Mr. Rawson • GCSE Computer Science



Year 10 2026

1.2 Quiz — Data Representation

Fifteen minutes. Books closed — no notes. Do not use a calculator. This is laid out like the real paper, so that the real paper holds no surprises. Hand it in when the time is up — I will mark it.

1 Computers represent data in binary form.

(a) Tick (✓) **one** box to identify the number of bits in one **nibble**.

- ☐ 2
☐ 4
☐ 8
☐ 16

[1]

(b) Tick (✓) **one** box to identify the largest denary number that can be stored in 8 bits.

- ☐ 128
☐ 254
☐ 255
☐ 256

[1]

(c) Complete the table by writing the missing denary, 8-bit binary or hexadecimal values.

Denary	8-bit binary	Hexadecimal
58		3A
153	10011001	
	11001110	CE
123		7B

[4]

(d) Add these two 8-bit binary numbers using binary addition.

Show your working out.

(I have given you a place-value grid — fill it in on your own.)

	128	64	32	16	8	4	2	1
	0	0	1	1	0	1	0	1
+	0	0	0	1	1	0	1	1
Answer								
Carries								

[2]

(e) Add these two 8-bit binary numbers using binary addition.

Show your working out.

	128	64	32	16	8	4	2	1
	1	1	0	1	1	1	0	0
+	0	1	0	0	1	1	0	0
Answer								
Carries								

[2]

(f) Describe what happens when the answer to part (e) is stored in 8 bits.

(This one is mine, not the exam’s — but the specification does ask you to explain it.)

.....
.....
..... [2]

(g) Show the result of a left binary shift of 1 place on the binary number 00100110.
 (Grid again — fill it in yourself.)

	128	64	32	16	8	4	2	1
	0	0	1	0	0	1	1	0
Answer								

[1]

(h) Tick (✓) **one** box to identify the result of a 2-place right binary shift on the binary number 01010100.

- ☐ 00010101
- ☐ 00101010
- ☐ 01010000
- ☐ 10101000

[1]

(i) Describe the binary shift that can be used to divide any number by 4.

.....

..... [2]

(j) The binary number 10110001 is stored in 8 bits.

(i) Identify the place value of the **most significant bit**.

..... [1]

(ii) Identify the place value of the **least significant bit**.

..... [1]